STATEMENT OF WORK

FOR

C40 & C41 RF ROOF REPLACEMENT

CONSTRUCTION

CAT: ☐A1 ☐ A2 ☒ A3

REVISION 1

DATED: August *28th, 2026*

PREPARED BY: Jaspreet Sidhu
Digitally signed by Jaspreet Sidhu
DN: C=US, E=jsidhu@pppl.gov, O=PPPL, OU=PMO, CN=Jaspreet Sidhu
Reason: I have reviewed this document
Date: 2026.09.16 13:22:19-04'00'
Cognizant Individual

REVIEWED BY: Kyle Jacobs
Digitally signed by Kyle Jacobs
DN: C=US, E=kjacobs@pppl.gov, O=PPPL, OU=F&SS, CN=Kyle Jacobs
Date: 2026.10.01 09:24:41-04'00'
Responsible Engineer

REVIEWED BY: Robert S. Oh
Digitally signed by Robert S. Oh
DN: C=US, E=roh@pppl.gov, O=PPPL, OU=Facilities and Site Services, CN=Robert S. Oh
Date: 2026.10.01 15:28:32-04'00'
Responsible Engineer

REVIEWED BY: Joy E. Fleming, MPH, CIH, CSP
Digitally signed by Joy E. Fleming, MPH, CIH, CSP
Date: 2026.10.02 10:15:30 -04'00'
ES&H

REVIEWED BY: Marlo Ramos
Digitally signed by Marlo Ramos
Date: 2026.10.02 10:38:35 -04'00'
Quality Assurance

APPROVED BY: Jermain Lewis
Digitally signed by Jermain Lewis
DN: C=US, E=jermain.lewis@gmail.com, O=PPPL - Design and Construction, OU=Project Managment Office, CN="Jermain Lewis "
Location: Fire Alarm
Reason: I attest to the accuracy and integrity of this document
Date: 2026.10.06 10:48:38-04'00'
Project Manager (PM)

PRINCETON UNIVERSITY
PLASMA PHYSICS LABORATORY
P.O. BOX 451
PRINCETON, N.J. 08543
609-243-2000

1.0 INTRODUCTION & SCOPE

1.1. INTRODUCTION

1.1.1. The Princeton Plasma Physics Laboratory (PPPL) is used to house, operate, and maintain advanced experimental equipment and infrastructure critical to the mission of the Laboratory.

1.1.2. There are several existing roofs (C40 RF Building & C41 CS Building) which are past their useful life requiring costly maintenance to upkeep. Severe water infiltration occurs regularly at failing joints, punctures, and flashing. Portions of existing metal coping have pulled away during storm events causing large sections of membrane to separate from the substrate. A catastrophic failure to one of these roofs would negatively impact the ability to conduct the top priority experimental operations in the Laboratory.

1.2. SCOPE

1.2.1. The Subcontractor shall provide all materials, labor, supervision, equipment, and expertise required to replace the noted roofs and specified components per the associated contract design documents. Construction scope includes the following:

1.2.1.1. Remove and replace existing built-up roofs at PPPL. Buildings include:

1.2.1.1.1. C40 – RF Building

1.2.1.1.2. C41 – CS Building

1.2.1.2. Demolition including removal of existing membrane, cover board, insulation, metal trim, flashing, coping and any other miscellaneous roofing material accessories as shown on the as-bid design document package.

1.2.1.3. Remove and recycle stone ballast on ballasted roofs where applicable. Recycle documentation to be provided to PPPL including final weights.

1.2.1.4. Additional integrated roof elements shall be inspected and replaced as specified within the as-bid design document package. Examples of these elements include: wood blocking, roof drains, gutters, scuppers, downspouts, and metal/concrete roof panels.

1.2.1.5. Abandoned equipment shall be removed to a point that does not affect new roof installation. Any equipment that should be temporarily relocated so roof replacement work can be completed shall be the responsibility of the roofing contractor as noted in the as-bid design document package.

1.2.1.6. Replacing of any ductwork/insulation and any other mechanical equipment or components shall also be included in the base scope price as noted in the as-bid design document package.

1.2.1.7. Contractor to furnish and install counter flashing and copings, expansion joints and other roofing components as required for manufacturer's warranty and as noted in the as-bid design document package.

1.2.1.8. Roof perimeter guardrails shall be provided on all roofs as noted in the as-bid design document package.

1.2.1.9. New and higher curbs as required due to the new roof installation work as noted in the as-bid design document package.

1.2.1.10. Contractor to build up existing roof vents as required due to the new roof work as noted in the as-bid design document package.

1.2.1.11. Contractor to replace all existing roof scuppers with new as noted in the as-bid design document package.

1.2.1.12. Contractor to replace all existing roof drains with new as noted in the as-bid design document package.

1.2.1.13. Contractor to temporary relocate and reinstall any fall protection anchor systems identified on roof sections.

1.2.1.14. Contractor to install secondary drainage per code in addition to existing drains for all roof sections.

1.2.1.15. If necessary, existing asbestos elements shall be mitigated by PPPL. Coordination of this work shall be completed in concert with all other construction activities. The subcontractor to follow directly after PPPL abatement contractor to install new roofing system. The construction subcontractor shall be responsible to coordinate all work with PPPL.

1.2.1.15.1. As noted in the as-bid design document package, the current detail calls for existing asbestos vertical panels to be cut and modified in the field to account for the new roof height. PPPL to coordinate this work separately.

1.2.1.15.2. Contractor to provide add alternate detail/pricing if there is an option to terminate new roof membrane without having to cut the existing asbestos panels.

1.2.1.15.2.1. PPPL to review/approve the add alternate detail if deemed safe and appropriate.

1.2.1.16. Any and all equipment that requires power disconnect and or relocation, has to be coordinated with PPPL a minimum of 4 weeks in advance to mitigate impact to lab operations. All equipment that will be out of service shall be minimized.

1.2.1.17. All roof fans and equipment are to be salvaged, protected, and reinstalled as noted in the as-bid design document package.

1.2.1.18. Contractor to provide and include a separate add alternate line item pricing in their bid for metal duct protector for areas where there is ductwork modifications/changes taking place as noted in the as-bid design document package.

2.0 APPLICABLE DOCUMENTS

2.1. Applicable documents and referenced standards shall follow the latest edition and/or revision unless noted otherwise. The applicable documents shall include, but are not limited to the following:

2.1.1. New Jersey Uniform Construction Code (NJUCC, NJAC 5:23) Adopted Sub-Codes:

2.1.1.1. OSHA, OCCUPATIONAL SAFETY AND HEALTH ACT

2.1.1.2. CODE OF FEDERAL REGULATION 29 CFR 1910, OCCUPATIONAL SAFETY AND HEALTH STANDARDS
2.1.1.3. CODE OF FEDERAL REGULATION 29 CFR 1926, CONSTRUCTION INDUSTRY REGULATIONS
2.1.1.4. ICC, INTERNATIONAL CODE COUNCIL
2.1.1.5. ICC FAMILY AND REFERENCED STANDARDS
2.1.1.6. 2021 IBC, INTERNATIONAL BUILDING CODE
2.1.1.7. 2021 IECC, INTERNATIONAL ENERGY CONSERVATION CODE
2.1.1.8. 2021 IEBC, INTERNATIONAL EXISTING BUILDING CODE
2.1.1.9. 2021 IFC, INTERNATIONAL FIRE CODE
2.1.1.10. 2021 IMC, INTERNATIONAL MECHANICAL CODE
2.1.1.11. 2021 IPC, INTERNATIONAL PLUMBING CODE
2.1.1.12. A117.1-2017 STANDARD FOR ACCESSIBLE AND USABLE BUILDINGS AND FACILITIES
2.1.1.13. NFPA, NATIONAL FIRE PROTECTION ASSOCIATION
2.1.1.14. 2021 NFPA 1, FIRE CODE
2.1.1.15. 2022 NFPA 13, STANDARD FOR THE INSTALLATION OF SPRINKLER SYSTEM
2.1.1.16. 2023 NFPA 70, NATIONAL ELECTRIC CODE
2.1.1.17. 2021 NFPA 70E, STANDARD FOR ELECTRICAL SAFETY IN THE WORKPLACE
2.1.1.18. 2022 NFPA 72, NATIONAL FIRE ALARM AND SIGNALING CODE
2.1.1.19. 2021 NFPA 101, LIFE SAFETY CODE
2.1.1.20. 2021 NFPA 5000, BUILDING CONSTRUCTION AND SAFETY CODE
2.1.1.21. ASHRAE, AMERICAN SOCIETY OF HEATING, REFRIGERATING, AND AIR-CONDITIONING ENGINEERS
2.1.1.22. 2017 ASHRAE 52.2, TESTING GENERAL VENTILATION AIR-CLEANING DEVICES
2.1.1.23. 2017 ASHRAE 55, THERMAL ENVIRONMENTAL CONDITIONS FOR HUMAN OCCUPANCY
2.1.1.24. 2019 ASHRAE 62.1, VENTILATION FOR ACCEPTABLE INDOOR AIR QUALITY
2.1.1.25. 2019 ASHRAE 90.1, ENERGY STANDARD FOR BUILDINGS
2.1.1.26. UNDERWRITER'S LABORATORY
2.1.1.27. FACTORY MUTUAL GLOBAL
2.1.1.28. ANSI, AMERICAN NATIONAL STANDARDS INSTITUTE
2.1.1.29. PPPL PROCEDURES AND POLICIES
2.1.1.30. PPPL ES&H 5008 (HTTP://WWW.PPPL.GOV/ESHIS/ESHD_MANUAL/SM.HTML)
2.1.1.31. 2016 DOE STANDARD 1066, FIRE PROTECTION
2.1.1.32. 2019 DOE ORDER 420.1C, FACILITY SAFETY
2.1.1.33. CODE OF FEDERAL REGULATION DOE 10 CFR 851, WORKER SAFETY AND HEALTH PROGRAM
2.1.1.34. DEPARTMENT OF ENERGY EXECUTIVE ORDER 13834: EFFICIENT FEDERAL OPERATIONS
2.1.1.35. ICC/ANSI A117.1-2009
2.1.1.36. AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
2.1.1.37. INTERNATIONAL ENERGY CONSERVATION CODE, 2021

2.1.1.38. AMERICANS WITH DISABILITIES ACT, 2010
2.1.1.39. ANSI/ASA S12.60; ACOUSTICAL PERFORMANCE CRITERIA, DESIGN REQUIREMENTS, AND GUIDELINES FOR SCHOOLS
2.1.1.40. ANSI/INFOCOMM 2M-2010, STANDARD GUIDE FOR AUDIOVISUAL SYSTEMS DESIGN AND COORDINATION PROCESS (PROJECT MANAGEMENT PROCESS)
2.1.1.41. ANSI/INFOCOMM 3M-2011 PROJECTED IMAGE SYSTEM CONTRAST RATIO
2.1.1.42. ANSI/INFOCOMM 4: 2012 AUDIOVISUAL SYSTEMS ENERGY MANAGEMENT
2.1.1.43. ANSI/INFOCOMM V202.01:2016, DISPLAY IMAGE SIZE FOR 2D CONTENT IN AUDIOVISUAL SYSTEMS
2.1.1.44. AV DESIGN REFERENCE MANUAL, FROM INFOCOMM INTERNATIONAL
2.1.1.45. AC 9000:2012 QUALITY MANAGEMENT SYSTEM CONDUIT, PATHWAY, AND DEDICATED EQUIPMENT ROOM CONSIDERATIONS
2.1.1.46. AUDIOVISUAL SYSTEM DESIGN AND COORDINATION COMPONENTS, FROM INFOCOMM INTERNATIONAL
2.1.1.47. ANSI/TIA-568.0-D – GENERIC TELECOMMUNICATIONS CABLING FOR CUSTOMER PREMISES, 2015
2.1.1.48. ANSI/TIA-568.1-D – COMMERCIAL BUILDING TELECOMMUNICATIONS CABLING STANDARD, 2015
2.1.1.49. ANSI/TIA 568-C.2 – BALANCED TWISTED-PAIR TELECOMMUNICATIONS CABLING AND COMPONENTS STANDARD, 2009
2.1.1.50. ANSI/TIA-568.3-D – OPTICAL FIBER CABLING COMPONENTS STANDARD, 2016
2.1.1.51. ANSI/TIA-568.4-D – BROADBAND COAXIAL CABLING AND COMPONENTS STANDARD, 2017
2.1.1.52. ANSI/TIA-569-D – TELECOMMUNICATIONS PATHWAYS AND SPACES, 2015
2.1.1.53. ANSI/TIA/EIA-598-D -OPTICAL FIBER CABLE COLOR CODING, 2014
2.1.1.54. ANSI/TIA-606-B – ADMINISTRATION STANDARD FOR COMMERCIAL TELECOMMUNICATIONS INFRASTRUCTURE, 2012
2.1.1.55. ANSI/TIA-607-C – COMMERCIAL BUILDING GROUNDING (EARTHING) AND BONDING REQUIREMENTS FOR TELECOMMUNICATIONS, 2015
2.1.1.56. DAVIS AND DAVIS, 1987 “SOUND SYSTEM ENGINEERING”
2.1.1.57. GIDDINGS, 1990, “AUDIO SYSTEMS DESIGN AND INSTALLATION”
2.1.1.58. INFOCOMM, AV/IT INFRASTRUCTURE GUIDELINES FOR HIGHER EDUCATION
2.1.1.59. NFPA 70 – NATIONAL ELECTRICAL CODE (NEC)
2.1.1.60. NFPA 72 – NATIONAL FIRE ALARM AND SIGNALING CODE
2.1.1.61. NFPA 75 – STANDARD FOR THE PROTECTION OF INFORMATION TECHNOLOGY EQUIPMENT
2.1.1.62. NFPA 76 – STANDARD FOR THE FIRE PROTECTION OF TELECOMMUNICATIONS FACILITIES
2.1.1.63. NIST 800-53, SC-15

2.1.2. PPPL SUBCONTRACTOR REQUIREMENTS MANUAL (MNL-007)

3.0 APPLICABLE DRAWINGS

3.1. As noted in the as-bid design document package

4.0 RESPONSIBILITIES

4.1. PRINCETON PLASMA PHYSICS LABORATORY (PPPL)

4.1.1. All construction activities must be coordinated with the Princeton Technical Representative (PTR) to minimize systems down time and shall not have any negative impact on experimental operations.

4.1.2. All communications on technical matters shall be directed to the assigned PTR

4.1.3. All communications on administrative matters shall be directed to the PPPL Procurement Representative.

4.1.4. PPPL permits all employees; both PPPL and subcontractor, to issue a "STOP WORK" Order, if unsafe work practices are observed.

4.2. SUBCONTRACTOR

4.2.1. The Subcontractor shall provide a single point of contact for any communication between PPPL and the Subcontractor.

4.2.2. In the event work has stopped, due to unsafe conditions or other related construction activity, the Subcontractor shall notify the Princeton Technical Representative immediately. See "Notification Requirements Off-Normal Events and Issues" attached.

4.2.3. The Subcontractor shall be responsible for coordinating all means and methods with PPPL prior to bid i.e. onsite material storage, laydown area, vehicle access, equipment usage, etc. Various restrictions shall be imposed by PPPL depending on the means and methods of the Subcontractor for specific activities. Conditions requiring temporary services (i.e. electric, water, HVAC, lighting, etc.) shall be provided by the Subcontractor. All change order costs associated with foreseen means and methods of the Subcontractor will not be accepted by PPPL.

4.2.4. The Subcontractor is responsible to verify and confirm the scope of work within this Statement of Work and referenced construction documents. All questions regarding the scope of work shall be received prior to bid.

4.2.5. Prior to bid, the Subcontractor shall identify any facility changes, modifications, or obstructions to be mitigated in order to safely and efficiently perform work.

4.2.6. Work shall be performed in accordance with 10 CFR 851 and OSHA requirements.

4.2.7. The Subcontractor shall dispose of all construction debris in accordance with Federal, State, and Local requirements. Subcontractor leased waste containers shall be provided. In an effort to reduce waste to landfill per DOE E.O. 13834, the Subcontractor shall perform an inventory of waste to be generated and put in process means to recycle and reuse materials generated. Inventory of materials generated for the purpose of diverting waste from landfill shall be

provided to the PTR. Additionally, the subcontractor shall provide the PTR data for all items recycled by type. Data shall be reported in tons and provided monthly.

4.2.8. All work should be completed during normal working hours of 7:00 AM to 3:30 PM, Monday through Friday. Off hours shall be communicated and approved with PPPL prior to the start of work. Noise restrictions may be placed on the contractor where off-hour work is necessary. The Subcontractor shall submit a construction schedule as part of their bid package. Revisions to this schedule shall be submitted as construction progresses.

4.2.9. Periodic status meetings shall be held by the Subcontractor with PPPL representatives present. Status meetings shall be held on a weekly basis unless otherwise agreed upon by the PTR. Meeting minutes delineating all action items shall be kept by the Subcontractor and distributed to all attendees.

4.2.10. The subcontractor shall not demolish and/or remove more existing roofing area than they can complete in a single day. If the subcontractor cannot meet this requirement, coordinate with PPPL prior to demolition.

4.2.11. Access to the property will only be granted to those subcontractors and lower tier subcontractors who present a REAL ID – compliant ID or a valid passport at the gate.

4.2.12. The Subcontractor shall notify and obtain PPPL written approval for any lower-tier Subcontractor proposed for this project, prior to their arrival on site.

4.2.13. All subcontractors and lower tier subcontractors who are not badged must provide the PTR their names, citizenship status and contact information 48 hours prior to arrival onsite.

4.2.14. The Subcontractor shall protect any and all infrastructure within the project limits to prevent damage during demolition and construction activities.

4.2.15. At the end of every workday, the subcontractor to leave the building roofs and/or work areas waterproof and protected from the weather.

4.2.16. Subcontractor to maintain a clean work area at all times. All trash is to be removed and work area broom cleaned daily. Any loose materials; tools and equipment are to be secured daily.

4.2.17. All Subcontractors shall be directed to park within the lower visitor's parking lot unless circumstantial situations arise.

5.0 REQUIREMENTS

5.1. PERFORMANCE REQUIREMENTS

5.1.1. SUSTAINABILITY

5.1.1.1. PPPL's prime contract requires the purchase of certain types of environmentally preferable products such as, but not limited to, EPEAT-registered electronic devices, ENERGYSTAR and FEMP-listed energy-consuming equipment, Water-Sense listed products, bio-based materials, EPA-designated recycled content products, non ozone-depleting chemicals and non-toxic or less toxic alternatives when practical. Where such products are available and meet the technical requirements of the work, they should be installed to the maximum extent practical. Additional information on environmentally

preferable products is available online at:
https://pppl.princeton.edu/PPPL_Environmentally_Preferable_Purchasing

6.0 TEST & INSPECTION REQUIREMENTS

6.1. PERFORMANCE TESTS

6.1.1. As noted in the as-bid design document package

6.2. ACCEPTANCE TESTS

6.2.1. Daily inspections of the job site, the adjacent areas, and protective systems shall be made by the Subcontractors Competent Person for evidence of a situation that could result in possible failure of protective systems or other hazardous conditions. Inspections shall be conducted by the Subcontractors Competent Person prior to the start of work and as needed during the shift. Daily inspection reports shall be documented by the Subcontractor and be made available to PPPL upon request.

6.2.2. A final walkthrough inspection will be conducted with the Subcontractor, the engineer of record, and PPPL representatives to evaluate the project construction upon completion of work. A punch list shall be created by the Subcontractor documenting all deficiencies and satisfactory resolutions agreed upon with PPPL.

6.2.3. All deficiencies not meeting the requirements of this statement of work shall be corrected at the Subcontractor's expense in a timely manner.

6.3. SUPPLIER HOLD POINTS

6.3.1. The subcontractor shall coordinate asbestos removal with PPPL as specified on SOW section 1.2.1.15.

6.3.2. The subcontractor shall coordinate removal of existing equipment with PPPL as specified on SOW section 1.2.1.5.

6.3.3. The subcontractor shall coordinate all roof work activities with PPPL in advance and shall provide temporary protection of all equipment within the specified buildings.

6.4. QUALITY CONTROL RECEIPT INSPECTIONS

6.4.1. As noted in the as-bid design document package

7.0 QUALIFICATIONS

7.1. Pre-approval is required before special purpose vehicles or equipment such as mobile cranes, forklifts, manlifts, etc. can be delivered and accepted onsite. The Subcontractor must submit to PPPL operator training qualifications, as well as equipment maintenance and inspection records for all special purpose equipment. Inspection records shall include model and serial number for each specific piece of equipment. In addition, the equipment manual shall be submitted for review and provided on site along with the equipment. Upon delivery and prior to use, PPPL reserves the right to perform equipment inspections to verify that the equipment is acceptable for usage onsite. For additional requirements not mentioned refer to PPPL Engineering Standards ES-MECH-007, ES-MECH-010, ES-

MECH-011, and ES-MECH-012. PPPL Engineering Standards may be found online at http://bp-pub.pppl.gov/EngStds.html.

7.2. For all hoisting and rigging, a Lift Plan shall be submitted to PPPL for approval 7 days prior to all activity, per PPPL Engineering Standards ES-MECH-007. The lift plan shall include:

7.2.1. Periodic equipment inspection reports

7.2.2. Monthly hook and wire rope inspection reports

7.2.3. Lift data sheet detailing loads, center of gravity, attachment points, component capacities, calculations, lift angle, and swing location. Requires signature by the Subcontractor's Lift Engineer.

7.2.4. Qualifications of the Rigger and Signal Person. An officer from the Subcontractor's firm shall sign that the stated Rigger and Signal Person are qualified per OSHA 1926.1400.

7.2.5. Equipment manual and load charts.

7.3. Subcontractor must provide any special construction equipment required to perform work. Subcontractor employees assigned to tasks must carry with them the basic tools of their trade.

8.0 ENVIRONMENT, SAFETY, AND HEALTH

8.1. SAFETY PROGRAM

8.1.1. All work shall comply with PPPL Subcontractor Requirements Manual (MNL-007).

8.1.2. Subcontractor must prepare a written Integrated Safety Management (ISM) Plan (construction project safety and health plan) that implements the requirements noted and will obtain approval of the plan by the PPPL Construction Manager prior to commencement of any work covered by the Plan. In the Plan, the Subcontractor must designate the individual(s) responsible for its on-site implementation, specify qualifications for those individuals, and provide a list of those project activities for which subsequent job hazard analyses are to be performed. The level of detail within the ISM Plan should be commensurate with the size, complexity and risk level of the work to be performed.

8.1.3. The Subcontractor shall identify for approval by the PPPL Construction Manager competent persons for activities as required by applicable OSHA standards. These should be included in the ISM plan and amended as necessary.

8.1.4. The Subcontractor shall ensure that workers not following safe work practices are subject to Subcontractor's disciplinary process. This policy shall be included within the submitted ISM plan.

8.1.5. The Subcontractor shall provide approval drawings and/or other documentation of protective measures for which applicable OSHA standards require preparation by a Professional Engineer or other qualified individual. These should be included in the ISM Plan and include but are not limited to Scaffolding and Fall Protection barriers and anchors.

8.2. COMPLIANCE

8.2.1. The Subcontractor must comply with:

8.2.1.1. The specific PPPL documents, requirements, permits and courses listed in this Statement of Work.

8.2.1.2. The completed Subcontractor ISM Plan.

8.2.1.3. All applicable Federal, State, and local laws, regulations and requirements whether or not they are specifically listed in the Statement of Work or subcontract.

8.2.2. All work shall be in accordance with all OSHA, DOE and PPPL requirements. Care shall be taken to protect personnel, employees and surroundings.

8.3. GENERAL NOTES

8.3.1. The Subcontractor's safety record will be considered in the bid evaluations.

8.3.2. Subcontractor's personnel working on-site at PPPL for more than 40 hours will be required to successfully complete PPPL's General Employee (safety) Training and receive a PPPL Identification Badge. PPPL Identification Badges are to be returned to PPPL upon completion of subcontract.

8.3.3. For each Job Order the Subcontractor shall prepare and have approved by the PPPL Construction Manager, a Job Hazard Analyses (JHAs) in accordance with PPPL procedure ESH-004 prior to commencement of affected work. The JHAs must identify foreseeable hazards and protective measures and address further hazards revealed by supplemental information provided by the PPPL Construction Manager. The JHA form can be found at the following web site http://bp.pppl.gov/JHA_online.doc

8.3.4. The Subcontractor shall ensure workers are aware of foreseeable hazards and the protective measures as defined by JHAs and require that workers acknowledge being so informed. This policy shall be included within the submitted ISM plan.

8.3.5. The Subcontractor shall maintain a designated representative onsite during all active construction that is knowledgeable of the associated hazards and has the authority to act on behalf of the Subcontractor and who makes frequent and regular inspections of the construction work site to identify and correct any instance of noncompliance with project safety and health requirements.

8.3.6. The Subcontractor shall instruct workers to report to the Construction Subcontractor's designated representative hazards not previously identified or evaluated. If immediate corrective action is not possible or the hazards fall outside the project scope, the Construction Subcontractor must immediately notify the affected workers, post appropriate warning signs, modify the appropriate JHAs, and implement needed interim control measures as well as notify the PPPL Construction Manager of the action taken. The Construction Subcontractor or the designated representative must stop work in the affected area until appropriate protective measures are established.

8.3.7. All Subcontractor vehicles and equipment must be in good working order with no leaks of any kind. Any spills must be reported immediately to PPPL and work stopped until spill is addressed. The Subcontractor is responsible for remediation costs caused by negligence and or faulty equipment. PPPL Environmental Services will determine the proper method of remediation.

8.3.8. PPPL has safety programs that include the following areas: Confined Space Permits, Digging Permits, Penetration and Fire Seal permits, Flame permits, Radiation Work permits, Lock-

out/Tagout of energy sources, and restrictions on working alone. The Subcontractor must confirm with their PPPL contact person that all prerequisites have been met before initiating any activity that involves these areas.

8.3.9. No burning, grinding or welding is permitted without a PPPL Hot Works permit.

8.3.10. All waste, including paints and other chemicals must be disposed of in accordance with local, state and Federal regulations and PPPL requirements.

8.3.11. All waste materials generated shall be disposed of according to the Department of Energy E.O. 13963 in an effort to divert waste to landfills.

8.3.12. Weight report for all materials recycled must be reported monthly to the PTR.

8.3.13. Safety data sheets for all paints, adhesives, chemicals, stains and solvents must be submitted and approved prior to the start of work.

8.3.14. Barricade tape, when needed, shall only be red "Danger" tape where personnel hazards exist. Yellow "Caution" tape may only be used for equipment protection. All barricade tape will be posted with the name and phone number of a contact. The PPPL representative may provide the contact information.

8.3.15. All requirements of this Statement of Work apply to all sub-tier contractors.

8.3.16. Subcontractors must have their employees trained and provide PPPL with their qualification prior to performing this work. PPPL ES&H training courses that are more specific to the Subcontractor's laboratory work scope may also be required. If PPPL determines that training of Subcontractor employees is inadequate, PPPL reserves the right to require Subcontractor to obtain certified training.

8.3.17. Silica, primarily from cutting, drilling, jackhammering, grinding, or milling of concrete materials requires compliance with OSHA 29 CFR 1926.1153(d), exposure monitoring, using an exposure limit of 25µg/m3 (American Conference of Governmental Industrial Hygienists Threshold Limit Value - ACGIH TLV) as opposed to the OSHA Permissible Exposure Limit of 50 µg/m3. The table in 29 CFR 1926.1153(c) may NOT be used. The contractor must perform exposure monitoring or provide enough data from similar work to demonstrate compliance.

8.3.18. Fall protection shall comply with OSHA 29 CFR 1926.500-503; however, the use of a safety monitoring system will NOT be permitted to be used as fall protection at PPPL. If a warning line system is to be used, guardrails or personal fall arrest systems are required for work outside the warning lines. A description of the fall protection methods to be used by the subcontractor must be included in the ISM Plan (Health and Safety Plan).

9.0 QUALITY ASSURANCE REQUIREMENTS

9.1. GENERAL NOTES

9.1.1. The Subcontractor shall establish and maintain an effective Quality Assurance Program. This program shall assure that the Subcontractor's work meets the required level of quality and is performed in accordance with this statement of work.

9.1.2. PPPL's review and/or approval of Subcontractor's documents nor PPPL's inspection of Subcontractor's items or services shall not relieve the Subcontractor of responsibility for full compliance with requirements of this statement of work.

9.1.3. Subcontractor shall perform work on only with tradesmen who have a demonstrated history of the quality of work required with the specified products on previous jobs.

9.1.4. The Subcontractor is responsible for inspecting all work in progress to ensure compliance with this Statement of Work. Daily inspection reports shall be documented by the Subcontractor and be made available to PPPL upon request.

9.1.5. PPPL written approval is required prior to use of any lower tier subcontractors. The Subcontractor is responsible for assuring that all requirements and restrictions are imposed on any lower tier contractors.

9.1.6. The assigned PPPL Construction Coordinator and the Subcontractors representative/foreman are required to hold daily job briefings. The briefings shall be held at the start of a shift, as schedule permits.

9.1.7. Project submittals shall be submitted by Contractor for PPPL approval in advance with ample time to review and approve. No work shall be completed without written approval by PPPL.

9.1.8. Materials:

9.1.8.1. Only new first quality materials will be supplied and used in this project.

9.1.8.2. Materials shall be delivered to the site in unopened, original containers, bearing the manufacturers printed label stating the quality, brand or trade name, batch number, date of manufacture and directions for use. Delivery of small day-to-day supplies will not be permitted.

9.1.8.3. PPPL & DOE have active programs to prevent use of Suspect & Counterfeit Items (SCI). Material(s) and/or product(s), including those components, parts, and materials that are permanently installed into assemblies furnished under this purchase order/subcontract shall be new. The Subcontractor shall verify conformance of purchased items to drawing and specification requirements and inspect items to detect signs of use or indications that raise questions of authenticity. Subcontractor shall actively prevent introduction of SCI to PPPL by inspection of all items brought on site to detect:

9.1.8.3.1. Remanufactured, rebuilt, or used parts represented as new.

9.1.8.3.2. Counterfeit parts (fraudulently labeled or marked with another manufacturer's name).

9.1.8.3.3. Misrepresented parts.

9.1.8.3.4. The Subcontractor shall assure that fasteners do not have headmarks listed on the DOE Headmark List (See Attachment #2 for DOE Head Mark List for high strength fasteners).

9.1.8.3.5. Many items are counterfeited, but fasteners and flanges are especially common. Flanges shall be inspected for required and expected markings.

9.1.8.3.6. Regardless of application, high strength bolts, defined as bolts graded to have a tensile strength greater than 100 ksi, must be from a domestic manufacturer, must comply with a national standard, must have the original manufacturer's Certified Material Test Reports (CMTR), and must be traceable to the CMTR(s) using lot numbers or other unique identification.

10.0 SHIPPING STORAGE AND HANDLING

10.1. Limited storage area will be available for materials and tools. The Subcontractor will be responsible for complying with the manufacturer's recommendations for protecting materials from damage while stored and during installation. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

10.2. Temporary storage shall be provided by the Subcontractor as needed.

11.0 WARRANTY

11.1. Upon completion of all roof replacement work, Contractor shall issue Manufacturer's 30 Year Warranty as well as a 2 Year Installers Warranty.

11.2. The Warranty coverage shall include incidental membrane punctures, damage from hail and a minimum of 72 MPH peak gust wind speeds.

12.0 ATTACHMENTS

12.1. Attachment #1 – Notification Requirements Off-Normal Events and Issues

12.2. Attachment #2 – DOE Headmark List

12.3. Attachment #3 – C40 & C41 As-Bid Design Document Package

13.0 DOCUMENTATION & DELIVERABLES

Deliverables List

- Please provide a comprehensive bid proposal for the roof replacement work, structured to clearly present pricing based on the following execution scenarios as laid out in the ad-bid design documents:
    - Bid Price #1: Phase A, Phase B and Phase C
        - The total cost to complete all three roof replacements as a single, unified project.
    - Bid Price #2: Individual Phased Pricing
        - The separate cost for each building's roof replacement, as if each phase were executed individually:
            - Phase A
            - Phase B
            - Phase C

PO / Subcontract / BOA / BPA #:_____

#	Physical Deliverables Required	When Deliverable	Deliverable Received

		Is Required	(✓)
1	Phase A/B/C Roof Replacement work as noted in the as-bid design document package.	PS/D	

Exceptions (Add justification for any missing physical deliverables that will not be received):

#	Document Deliverables Required	When Deliverable Is Required	Deliverable format (paper, electronic etc.)	Storage Location for Deliverable	Deliverable Received (✓)
1	Construction and Solid Waste Materials Recycled Data (as req'd)	D	P,E	Ops Center	
2	Construction Schedule	N/D	P,E	Ops Center	
3	Meeting Minutes (as req'd)	D	P,E	Ops Center	
4	List of Lower Tier Subcontractors (as req'd)	PS/D	P,E	Ops Center	
5	Certified Material Test Reports (as req'd)	PS/D	P,E	Ops Center	
6	Daily Inspection Reports (as req'd)	D	P,E	Ops Center	
7	Punch List	D	P,E	Ops Center	
8	Equipment model, serial number, and manual (as req'd)	PS/D	P,E	Ops Center	
9	Equipment Maintenance and Inspection Records (as req'd)	PS/D	P,E	Ops Center	
10	Equipment Operator Training Qualifications or Certifications (as req'd)	PS/D	P,E	Ops Center	
11	Lift Plan (as req'd)	PS/D	P,E	Ops Center	
12	Installer Qualifications	N	P,E	Ops Center	
13	Integrated Safety Management Plan (ISM)	N	P,E	Ops Center	
14	Job Hazard Analysis (JHA)	PS/D	P,E	Ops Center	
15	PPPL Permits (as req'd)	PS/D	P,E	Ops Center	
16	Safety Data Sheets (as req'd)	PS/D	P,E	Ops Center	

17	Training Qualifications	PS/D	P,E	Ops Center	
18	Quality Plan	PS/D	P,E	Ops Center	
19	Project Submittals (Contract Documents)	PS/D	P,E	Ops Center	
Exceptions (Add justification for any missing document deliverables that will not be received):					

**Legend*

N = Notice to Proceed Requirement

PS = Prior to Start of Construction

D = During Construction/As Required

C = Project Completion

P,E = Paper or Electronic

Princeton Technical Representative/COG:_____

(Sign-off and provide to the Operations Center when job is completed and deliverables are dispositioned and placed/filed in Operations Center (or other Project, Department or Division designated file center)

Attachment #1

Notification Requirements Off-Normal Events and Issues

Notification Requirements Off-Normal Events and Issues

Note: Contacting the PPPL Emergency Services Unit at X3333 to obtain emergency assistance and taking immediate actions to protect workers, guests and visitors takes precedence over the notification requirements described herein.

The Subcontractor shall notify PPPL within 15 minutes of becoming aware of off-normal events and issues; off-normal events and issues include:

- Personnel injuries of any kind
- Near misses that could have resulted in significant (recordable injury or worse) worker injuries
- Uncontrolled or unforeseen personnel exposure to chemical, biological or physical hazards (e.g., noise, laser, ultraviolet light, heat, etc.)
- Fire emergency of any kind
- Any unexpected discovery of an uncontrolled hazardous energy source (e.g., live electrical power circuit, etc.) not including discoveries made by zero-energy checks and other precautionary investigations made before work is authorized to begin
- Any failure to follow a prescribed hazardous energy control process (e.g., lockout/tagout)
- Damage of any kind to Laboratory property
- Damage of any kind to personal property owned by PPPL employees, guests and visitors
- Any spills of chemicals, fuels, lubricants, etc.
- Other off-normal events and issues deemed by the Subcontractor to require notification

In the event of an off-normal event or issue, the Subcontractor and its lower-tier subcontractors

SHALL:

- Notify PPPL Emergency Services Unit at X3333 for emergency response, if required
- Take immediate action to evacuate and post affected area, if required
- Cease related work or work in affected area
- Take actions to post and safe-off affected area
- Within 15 minutes: Contact Subcontractor Site Supervisor, PPPL PTR, ESH Construction Safety Engineer and ESH Head

Regarding these off-normal events or issues, no recovery actions (other than those immediately required to preserve life safety) or repairs are to take place until they are reviewed by the Subcontractor and PPPL, and specifically authorized by the PPPL PTR or ESH Department Head. Resumption of work that has been stopped by PPPL due to off-normal events and issues must follow the provisions of PPPL Policy P-012.

Attachment #2

DOE Headmark List

DOE Headmark List

ANY BOLT ON THIS LIST SHOULD BE TREATED AS DEFECTIVE WITHOUT FURTHER TESTING.



ALL GRADE 5 AND GRADE 8 FASTENERS OF FOREIGN ORIGIN WHICH DO NOT BEAR ANY MANUFACTURERS' HEADMARKS:

 **GRADE 5**  **GRADE 8**

GRADE 5 FASTENERS WITH THE FOLLOWING MANUFACTURERS' HEADMARKS:

	MARK	MANUFACTURER		MARK	MANUFACTURER
	J	Jinn Her (TW)		KS	Kosaka Kogyo (JP)

GRADE 8 FASTENERS WITH THE FOLLOWING MANUFACTURES' HEADMARKS:

	MARK	MANUFACTURER		MARK	MANUFACTURER
	A	Asahi Mfg (JP)		KS	Kosaka Kogyo (JP)
	NF	Nippon Fasteners (JP)		RT	Takai Ltd (JP)
	H	Hinomoto Metal (JP)		FM	Fastener Co of Japan (JP)
	M	Minamida Sleybo (JP)		KY	Kyoei Mfg (JP)
	MS	Minato Kogyo (JP)		J	Jinn Her (TW)
	Hollow Triangle	Infasca (CA TW JP YU) (Greater than 1/2 inch dia.)			
	E	Dalai (JP)		UNV	Unytite (JP)

GRADE 8.2 FASTENERS WITH THE FOLLOWING HEADMARKS:

	MARK	MANUFACTURER
	KS	Kosaka Kogyo (JP)

GRADE A325 FASTENERS (BENNETT DENVER TARGET ONLY) WITH THE FOLLOWING HEADMARKS:

	MARK	MANUFACTURER
Type 1 	A325 KS	Kosaka Kogyo (JP)
Type 2 		
Type 3 		

Key: CA-Canada, JP-Japan, TW-Taiwan, YU-Yugoslavia

Attachment #3

C40 & C41 As-Bid Design Document Package